

**Lineare Algebra**  
**Serie 21**  
**Lukas Mengis**

## Exercise 1

Let  $K$  be a field. Let

$$p(X) = \sum_{i=0}^m c_i X^i \in K[X]$$

with  $c_i \in K$ . Let  $A, Q \in M_{n \times n}(K)$  be matrices such that  $Q$  is invertible.

$$\begin{aligned} p(QAQ^{-1}) &= \sum_{i=0}^m c_i (QAQ^{-1})^i \\ &= \sum_{i=0}^m c_i Q A^i Q^{-1} \\ &= Q \left( \sum_{i=0}^m c_i A^i \right) Q^{-1} \\ &= Q p(A) Q^{-1}. \end{aligned}$$

□

## Exercise 2

Let

$$p(X) = \sum_{i=0}^m c_i X^i \in K[X], p_k(X) = \sum_{i=0}^m c_i^{(k)} X^i \in \mathbb{C}[X]$$

with  $c_i \in \mathbb{C}$  and  $p_k \rightarrow p$  as  $k \rightarrow \infty$ . Let  $A, A_k \in M_{n \times n}$  with  $A_k \rightarrow A$  as  $k \rightarrow \infty$ .

$$p_k(A_k) = \sum_{i=0}^d c_i^{(k)} A_k^i$$

Fix  $i$ . Since  $c_i^{(k)} \rightarrow c_i$ ,  $A_k \rightarrow A$  as  $k \rightarrow \infty$  and Matrix multiplication is continuous,  $c_i^{(k)} A_k^i \rightarrow c_i A^i$ . Since this holds  $\forall i \leq d$  it follows:

$$p_k(A_k) \rightarrow p(A)$$

as  $k \rightarrow \infty$ .

□

## Exercise 3

Let  $\varepsilon > 0$  and  $A \in M_{n \times n}(\mathbb{C})$ .

$A$  is triangularisable  $\Rightarrow$  Hence  $\exists Q \in M_{n \times n}$  such that

$$Q A Q^{-1} = T$$

Where  $T$  is upper triangular.

Chose  $\delta_i \in \mathbb{C}$  such that  $|\delta_i| < \varepsilon$  and  $t_{ii} + \delta_i$  are pairwise distinct.

Then the Matrix

$$S = (b_{ij})_{ij} + (s_{ij})_{ij}$$

with

$$s_{ij} = \begin{cases} t_{ij} & \text{if } i \neq j \\ t_{ii} + \delta_i & \text{if } i = j \end{cases}$$

has pairwise distinct *eigenvalues* and is therefore diagonalizable.

Set

$$B = QSQ^{-1}$$

Since  $|\delta_i|$  is arbitrary small it follows that  $B$  can be arbitrary close to  $A$ . Furthermore,  $\delta_i$  can be chosen such that:

$$|a_{ij} - b_{ij}| < \varepsilon, \forall i, j.$$

□